9

EXAMPLES OF THEORIES OF PRESHEAF TYPE

In this chapter we discuss some classical as well as new examples of theories of presheaf type from the perspective of the theory developed in the previous chapters. We revisit in particular well-known examples of theories of presheaf type whose finitely presentable models are all finite, notably including the geometric theory of finite sets, and give fully constructive proofs of the fact that Moerdijk's theory of abstract circles and Johnstone's theory of Diers fields are of presheaf type. Next, we introduce new examples of theories of presheaf type, including the theory of algebraic extensions of a given field, the theory of locally finite groups, the theory of vector spaces with linear independence predicates and the theory of abelian ℓ-groups with strong unit. We also show that the injectivization of the algebraic theory of groups is not of presheaf type and explicitly describe a presheaf completion for it (in the sense of section 7.1.3).

9.1 Theories whose finitely presentable models are finite

Geometric theories whose finitely presentable models are all finite often happen to be of presheaf type. Corollary 8.2.4 provides a theoretical justification for this fact, while Corollary 9.1.3 below brings alternative evidence in this respect.

The following proposition shows that set-theoretic finiteness implies finite presentability in categories of set-based models of geometric theories over finite signatures. Recall that a model of a geometric theory over a finite signature Σ is said to be finite if the disjoint union of the MA for A a sort of Σ is a finite set.

Proposition 9.1.1 *Let $\mathbb{T}$ be a geometric theory over a finite signature. Then every finite model M of $\mathbb{T}$ is finitely presented (and hence finitely presentable).*

Proof For any $\mathbb{T}$-models M and N in **Set**, a bunch of functions $\xi_A : MA \to NA$ indexed by the sorts A of Σ is the set of components of a $\mathbb{T}$-model homomorphism $M \to N$ if and only if the following conditions are satisfied: (1) for any function symbol $f : A_1 \cdots A_n \to B$ of Σ and any tuple $(x_1, \ldots, x_n) \in MA_1 \times \cdots \times MA_n$, if $y = Mf(x_1, \ldots, x_n)$ then $\xi_B(y) = Nf(\xi_{A_1}(x_1), \ldots, \xi_{A_n}(x_n))$ and (2) for any relation symbol $R \rightarrowtail A_1 \cdots A_n$ of Σ and any $(x_1, \ldots, x_n) \in MR$, $(\xi_{A_1}(x_1), \ldots, \xi_{A_n}(x_n)) \in NR$.

Given a finite $\mathbb{T}$-model M, let us enumerate its elements (i.e. the elements of MA for some A) $a_1, \ldots, a_n$ in such a way that $a_i \neq a_j$ for $i \neq j$. Let us define a presentation of M as a model of $\mathbb{T}$ as follows: the generators $(x_1, \ldots, x_n)$ are the

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elements $(a_1, \ldots, a_n)$ of M, while the relations are given by the formulae which are either of the form $x_i = f(x_j, \ldots, x_k)$ (for any tuple $(a_i, a_j, \ldots, a_k)$ of elements of M and function symbol f of the signature of $\mathbb{T}$ such that $a_i = Mf(a_j, \ldots, a_k)$) or of the form $R(x_j, \ldots, x_k)$ (for any tuple $(a_j, \ldots, a_k)$ of elements of M such that $(a_j, \ldots, a_k) \in MR$). By the above remarks, this is indeed a presentation for M, and it is finite since the signature of $\mathbb{T}$ is finite. □

Remark 9.1.2 Proposition 9.1.1 specializes to Example 1.2 (4) [1] in the case where the signature of $\mathbb{T}$ contains no function symbols.

Proposition 9.1.1, combined with Theorem 8.2.10, yields the following result:

Corollary 9.1.3 *Let Σ be a finite signature and $\mathcal{A}$ a category of finite Σ-structures and Σ-structure homomorphisms between them. Then the geometric theory consisting of all the geometric sequents over Σ which are valid in all the structures in $\mathcal{A}$ (that is, the $\mathcal{A}$-completion of the empty theory $\mathbb{O}_\Sigma$ over Σ in the sense of Theorem 8.2.10) is of presheaf type, classified by the topos $[\mathcal{A}, \mathbf{Set}]$, and its finitely presentable models are precisely the finite ones.*

Remark 9.1.4 Corollary 9.1.3 can be notably applied to theories $\mathbb{T}$ with enough set-based models such that every model of $\mathbb{T}$ can be expressed as a filtered colimit of finite models of $\mathbb{T}$. Recall that, assuming the axiom of choice, every coherent theory has enough set-based models; it thus follows from the corollary that, assuming the axiom of choice, the injectivization of any finitary theory of presheaf type whose finitely presentable models are all finite is again of presheaf type.

Let us now discuss a few concrete examples of theories of presheaf type whose finitely presentable models are finite.

As an application of Corollary 8.2.4, we can recover at once the well-known results that the following theories are of presheaf type:

- The theory of decidable objects (cf. [56] and p. 907 of [55]);
- The theory of decidable Boolean algebras (cf. Example D3.4.12 [55]);
- The theory of linear orders;
- The theory of intervals (cf. Definition 2.1.27 and Proposition 2.1.28).

The fact that these theories are of presheaf type also follows from Corollary 9.1.3 by assuming the axiom of choice.

Notice that the former two theories are injectivizations of two cartesian theories, namely the empty theory over a signature consisting of just one sort and the algebraic theory of Boolean algebras. The latter two theories satisfy the hypotheses of Corollary 7.2.43, whence their injectivizations are of presheaf type too.

9.2 The theory of abstract circles

The *theory* $\mathbb{C}$ *of abstract circles* has been introduced by I. Moerdijk and shown in [66] to have the property that the points of *Connes' topos* of cyclic sets (introduced in [35]) can be identified with the set-based models of $\mathbb{C}$; in [66] it is also stated that Connes' topos actually classifies $\mathbb{C}$, but the argument given therein appears incomplete.

We shall prove in this section, by using Corollary 8.2.1, that $\mathbb{C}$ is of presheaf type classified by Connes' topos. Specifically, we shall show that $\mathbb{C}$ satisfies the hypotheses of the corollary with respect to its Horn part (i.e. the Horn theory consisting of the collection of all Horn sequents which are provable in $\mathbb{C}$). We will also prove that the injectivization of $\mathbb{C}$ is of presheaf type as well.

The signature Σ of the theory $\mathbb{C}$ consists of two sorts P and S (variables of type P will be denoted by letters $x, y, \ldots$, while variables of type S will be denoted by letters $a, b, \ldots$), two unary function symbols $0 : P \to S$ and $1 : P \to S$, two unary function symbols $\delta_0 : S \to P$ and $\delta_1 : S \to P$, one unary function symbol $* : S \to S$ and a ternary predicate R of type S. The individuals of sort P should be thought of as *points* and those of sort S as *segments*; the predicate R formalizes the partial operation of *concatenation* of segments.

The axioms of $\mathbb{C}$ can be formulated as follows (using the terminology of [66]):

1. *Non-triviality axioms*:

$$(\top \vdash_{[]} (\exists x)(x = x));$$
$$(\top \vdash_{x,y} (\exists a)(\delta_0(a) = x \wedge \delta_1(a) = y));$$
$$(0(x) = 1(x) \vdash_x \bot);$$

2. *'Equational' axioms*:

$$(\top \vdash_a a^{**} = a);$$
$$(\top \vdash_a \delta_0(a^*) = \delta_1(a));$$
$$(\top \vdash_x \delta_0(0(x)) = x \wedge \delta_1(0(x)) = x);$$
$$(\top \vdash_x 0(x)^* = 1(x));$$
$$(\delta_0(a) = x \wedge \delta_1(a) = x \vdash_{x,a} a = 0(x) \vee a = 1(x));$$

3. *Axioms for concatenation*:

$$(R(a, b, c) \wedge R(a, b, c') \vdash_{a,b,c,c'} c = c');$$
$$(R(a, b, c) \vdash_{a,b,c} \delta_0(c) = \delta_0(a) \wedge \delta_1(c) = \delta_1(b));$$
$$(R(a, b, c) \vdash_{a,b,c} R(c^*, a, b^*));$$
$$(R(a, b, d) \wedge R(d, c, e) \vdash_{a,b,c,d,e} (\exists e')(R(b, c, e') \wedge R(a, e', e)));$$
$$(R(a, b, 0(x)) \vdash_{a,b,x} a = 0(x));$$
$$(\delta_0(a) = x \vdash_{a,x} R(0(x), a, a);$$
$$(\delta_1(a) = \delta_0(b) \vdash_{a,b} (\exists c)R(a, b, c) \vee (\exists d)R(b^*, a^*, d)).$$

A model of $\mathbb{C}$ is said to be an *abstract circle*. Any set of points P of the circle S^1 defines an abstract circle S_P whose segments $a \in S$ such that $\delta_0(a) = x$ and $\delta_1(a) = y$ are the oriented arcs on S^1 from the point x to the point y. For any natural number $n > 0$ there is exactly one abstract circle, up to isomorphism, whose set of points has n elements; we shall denote it by the symbol C_n.

To prove that $\mathbb{C}$ is of presheaf type, we first notice that the sequent

$$(\delta_0(a) = \delta_0(b) \wedge \delta_1(a) = \delta_1(b) \vdash_{a,b} a = b)$$

is provable in $\mathbb{C}$. This sequent can be easily deduced as a consequence of the seventh axiom of group 3 and the fifth axiom of group 2. We shall refer to it as the 'uniqueness axiom'.

It follows that, modulo the non-triviality axioms and the uniqueness axiom, we can replace any expression (equivalent to one) of the form $(\exists c)(\phi(c) \wedge \delta_0(c) = x \wedge \delta_1(c) = y)$ arising in an axiom of $\mathbb{C}$ with the requirement that the unique segment d such that $\delta_0(d) = x$ and $\delta_1(d) = y$ satisfies ϕ. In particular, the fourth axiom of group 3 is provably equivalent, modulo the non-triviality axioms and the uniqueness axiom, to the following sequent:

$$\begin{aligned}(\delta_0(u) = \delta_0(b) \wedge \delta_1(u) = \delta_1(c) \wedge R(a,b,d) \wedge \\ \wedge R(d,c,e) \vdash_{a,b,c,d,e,u} R(b,c,u) \wedge R(a,u,e)).\end{aligned}$$

Similarly, the seventh axiom of group 3 is provably equivalent, modulo the non-triviality axioms and the uniqueness axiom, to the following sequent:

$$(\delta_0(c) = \delta_0(a) \wedge \delta_1(c) = \delta_1(b) \wedge \delta_1(a) = \delta_0(b) \vdash_{a,b,c} R(a,b,c) \vee R(b^*,a^*,c^*)).$$

From these remarks it follows that $\mathbb{C}$ admits a presentation in which none of the axioms involve existential quantifications except for the first and second of group 1.

Let us show that $\mathbb{C}$ satisfies the hypotheses of Corollary 8.2.1 with respect to its Horn part and the category of finite $\mathbb{C}$-models.

Given a homomorphism $f : c \to \mathrm{Hom}_{\mathcal{E}}(E, M)$, where c is a finite model of the Horn part of $\mathbb{C}$, M is a model of $\mathbb{C}$ in a Grothendieck topos $\mathcal{E}$ and E is an object of $\mathcal{E}$, we can 'localize' f (in the sense of section 8.2.1) a finite number of times (once for the first axiom of group 1 and once for each pair of points of c for the second axiom of group 1) to obtain Σ-substructure homomorphisms $f_i : c_i \hookrightarrow \mathrm{Hom}_{\mathcal{E}}(E_i, M)$ such that the structures c_i are finite models of the Horn part of $\mathbb{C}$ satisfying the non-triviality axioms (notice that any substructure of a structure of the form $\mathrm{Hom}_{\mathcal{E}}(E_i, M)$, where M is a model of $\mathbb{C}$ in $\mathcal{E}$, satisfies all the Horn sequents provable in $\mathbb{C}$). We can clearly further 'localize' each of these homomorphisms so as to obtain the satisfaction of all the other axioms of $\mathbb{C}$; since this can be done without modifying their domains, the final result will be a family of homomorphisms whose domains are structures satisfying all the axioms of $\mathbb{C}$.

Next, we notice that for any set-based model M of the Horn part of $\mathbb{C}$, any point $x \in MP$ and any segment $a \in MS$, there exists a finite substructure N of M such that NP contains x and NS contains a (take N equal to the substructure of M given by the sets

$$NP = \{x, \delta_0(a), \delta_1(a)\}$$

and

$$NS = \{a, 0(x), 1(x), 0(\delta_0(a)), 0(\delta_1(a)), 1(\delta_0(a)), 1(\delta_1(a)))\}).$$

Moreover, any two finite substructures N_1 and N_2 of M are contained in a common substructure N of M (take N equal to the substructure of M given by $NP = N_1P \cup N_2P$ and $NS = N_1S \cup N_2S$). This, combined with the above argument (specialized to the case $\mathcal{E} = \mathbf{Set}$), shows that every set-based model of $\mathbb{C}$ is a directed union of finite models of $\mathbb{C}$. It follows in particular that every finitely presentable $\mathbb{C}$-model is finite (it being a retract of a finite model). On the other hand, every finite model of $\mathbb{C}$ is finitely presentable as a model of the Horn part of $\mathbb{C}$ (cf. Proposition 9.1.1). We can thus conclude that the hypotheses of Corollary 8.2.1 are satisfied, whence $\mathbb{C}$ is of presheaf type classified by the topos of covariant set-valued functors on the category of finite models of $\mathbb{C}$.

Let us now show that the injectivization $\mathbb{C}_m$ of $\mathbb{C}$ is of presheaf type as well.

Due to the presence of conjunctions in the premises of some axioms of $\mathbb{C}$, we cannot directly apply Corollary 7.2.43 to conclude that $\mathbb{C}_m$ is of presheaf type. We shall instead apply Corollary 7.2.41. Since every $\mathbb{C}$-model in **Set** is a directed union of finite $\mathbb{C}$-models, the finitely presentable $\mathbb{C}_m$-models are exactly the finite $\mathbb{C}$-models. Moreover, by Proposition 7.2.31, the monic $\mathbb{C}$-model homomorphisms in **Set** are precisely the homomorphisms which are sortwise injective; indeed, the formulae $\{x^P . \top\}$ and $\{x^S . \top\}$ (strongly) present respectively the $\mathbb{C}$-models C_1 and C_2. To show that the hypotheses of Corollary 7.2.41 are satisfied, it remains to verify that for every Grothendieck topos $\mathcal{E}$, object E of $\mathcal{E}$ and Σ-structure homomorphism $x : c \to \mathrm{Hom}_{\mathcal{E}}(E, M)$, where c is a finite $\mathbb{C}$-model and M is a sortwise decidable $\mathbb{C}$-model, there exist an epimorphic family $\{e_i : E_i \to E \mid i \in I\}$ in $\mathcal{E}$ and for each $i \in I$ a $\mathbb{C}$-model homomorphism $f_i : c \to c_i$ of finite $\mathbb{C}$-models and a sortwise disjunctive Σ-structure homomorphism $x_i : c_i \to \mathrm{Hom}_{\mathcal{E}}(E_i, M)$ (in the sense of Lemma 7.2.16) such that $x_i \circ f_i = \mathrm{Hom}_{\mathcal{E}}(e_i, M) \circ x$. To this end, we make $\mathbb{C}$ into a one-sorted theory by identifying points x with the segments 0_x and rewriting its axioms appropriately, so that we can apply Proposition 7.2.39. The proposition yields an epimorphic family $\{e_i : E_i \to E \mid i \in I\}$ in $\mathcal{E}$ and for each $i \in I$ a surjective homomorphism $q_i : c \to c_i$, where c_i is a finite Σ-structure, and a disjunctive Σ-structure homomorphism $J_i : c_i \rightarrowtail \mathrm{Hom}_{\mathcal{E}}(E_i, M)$ such that $J_i \circ q_i = \mathrm{Hom}_{\mathcal{E}}(e_i, M) \circ f$. Now, since c is a $\mathbb{C}$-model and q_i is surjective, the structure c_i satisfies the non-triviality axioms. We can clearly suppose that $E_i \not\cong 0$ without loss of generality, and hence that the arrows J_i are injective. If we consider the image factorizations of the Σ-structure homomorphisms J_i (in the sense of Lemma 7.2.37), we thus obtain substructures $c_i' \rightarrowtail \mathrm{Hom}_{\mathcal{E}}(E_i, M)$ whose underlying sets are isomorphic to those of the c_i (since the J_i are injective) and Σ-structure homomorphisms $q'_i : c \to c'_i$. Since the c'_i have underlying sets isomorphic to those of the c_i, they all satisfy the non-triviality axiom, and, at the cost of refining the epimorphic family $\{e_i : E_i \to E \mid i \in I\}$, we can suppose that they satisfy all the other axioms of $\mathbb{C}$ (cf. the argument given above for showing that $\mathbb{C}$ satisfies the hypotheses of Corollary 8.2.1 with respect to its Horn part and the category of finite $\mathbb{C}$-models). So the hypotheses of Corollary 7.2.41 are satisfied, and we can conclude that $\mathbb{C}_m$ is of presheaf type classified by the topos of covariant set-valued functors on the category of finite models of $\mathbb{C}$ and injective homomorphisms between them.

Remark 9.2.1 In [33] we have described another theory, written over the signature of oriented groupoids (i.e. groupoids with a positivity predicate on arrows which is preserved by composition), which is classified by the cyclic topos, as well as a related theory classified by *Connes-Consani's epicyclic topos* (cf. section 4 of [37]). The advantage of working with oriented groupoids instead of abstract circles is that the composition of two segments (i.e. arrows of the groupoid) is always defined provided that the codomain of the first coincides with the domain of the second.

9.3 The geometric theory of finite sets

In this section we shall revisit, from the point of view of the theory developed in section 8.2.1, a well-known example of a theory of presheaf type, namely the geometric theory $\mathbb{T}$ of finite sets. Recall from Example D1.1.7(k) [55] that the signature Σ of $\mathbb{T}$ consists of one sort A and an n-ary relation symbol R_n for each $n > 0$. The axioms of $\mathbb{T}$ are as follows. For each n, we have the axiom

$$\sigma_n \equiv \left(R_n(x_1, \ldots, x_n) \vdash_{x_1,\ldots,x_n,y} \bigvee_{1\le i\le n} y = x_i \right),$$

expressing the requirement that if an n-tuple of individuals satisfies the relation R_n then it exhausts the members of (the set interpreting) the sort A. (The case $n = 0$ of this axiom is $(R_0 \vdash_{[]} \bot)$, which says that if R_0 holds then the interpretation of the sort A must be empty.) We also have the axiom

$$\left(\top \vdash_{[]} \bigvee_{n\in\mathbb{N}} (\exists x_1)\cdots(\exists x_n) R_n(x_1, \ldots, x_n) \right).$$

Finally, to ensure that the interpretations of the R_n are uniquely determined by that of the sort A (i.e. that R_n holds for all the n-tuples which exhaust the elements of the interpretation of the sort A, and not just for some of them), one adds the axiom

$$(R_n(x_1, \ldots, x_n) \vdash_{x_1\ldots,x_n} R_m(x_{f(1)}, \ldots, x_{f(m)}))$$

for any surjection $f : \{1, 2, \ldots, m\} \to \{1, 2, \ldots, n\}$, and

$$(R_n(x_1, \ldots, x_n) \wedge x_i = x_j \vdash_{x_1 \ldots, x_n} R_{n-1}(x_1, \ldots, x_{i-1}, x_{i+1}, \ldots, x_n))$$

whenever $1 \leq i < j \leq n$.

We can deduce that $\mathbb{T}$ is of presheaf type as an application of Corollary 8.2.1.

Notice that the set-based models of $\mathbb{T}$ can be identified with the finite sets, while the $\mathbb{T}$-model homomorphisms are precisely the surjective functions between them.

Let us regard $\mathbb{T}$ as a quotient of its Horn part (i.e. the theory $\mathbb{T}_H$ consisting of all the Horn sequents over the signature of $\mathbb{T}$ which are provable in $\mathbb{T}$). The criterion for finite presentability given by Lemma 6.1.12 ensures that every finite set, regarded as a model of $\mathbb{T}$, is finitely presentable as a model of $\mathbb{T}_H$; indeed, the finiteness of the structure immediately implies that the second condition of the lemma is satisfied, while the satisfaction of the first condition follows from the fact that every function from a finite $\mathbb{T}_H$-model of cardinality n to a set-based $\mathbb{T}_H$-model M which preserves the predicate R_n preserves the predicate R_m for any $m > n$ (by the 'introduction' and 'elimination' rules expressed by the last two groups of axioms of $\mathbb{T}$).

To apply Corollary 8.2.1, it thus remains to prove that the second condition in the statement of the corollary is satisfied. Given a homomorphism $f : a \to \mathrm{Hom}_{\mathcal{E}}(E, M)$, where a is a finite model $\mathbb{T}_H$ and M is a $\mathbb{T}$-model in a Grothendieck topos $\mathcal{E}$, if the cardinality of aA is n then for every $m > n$ the sequent σ_m is satisfied in a provided that σ_n is. Indeed, if $m \geq n$ then for any m-tuple $(y_1, \ldots, y_m)$ of elements of aA there exists a sub-n-tuple of elements of aA obtained by removing $m - n$ repetitions, which satisfies the relation R_n if the m-tuple $(y_1, \ldots, y_m)$ satisfies the relation R_m, since a, as a model of $\mathbb{T}_H$, satisfies the Horn sequent expressing the 'elimination' rule (i.e. the last group of axioms of $\mathbb{T}$). Similarly, by invoking the 'introduction' rules (i.e. the last but one group of axioms of $\mathbb{T}$), one can prove that if the cardinality of aA is n then for every $k < n$ the sequent σ_k is satisfied in a if σ_n is.

Let us show that we can inductively 'localize' (in the sense of section 8.2.1) the homomorphism f to eventually arrive at Σ-structure homomorphisms $f_i : a_i \to \mathrm{Hom}_{\mathcal{E}}(E_i, M)$ defined on Σ-structures a_i which satisfy all the axioms of $\mathbb{T}$.

Starting from a Σ-structure homomorphism $f : a \to \mathrm{Hom}_{\mathcal{E}}(E, M)$, where a is a finite $\mathbb{T}_H$-model, from the fact that M is a model of $\mathbb{T}$ it follows that there exist an epimorphic family $\{e_i : E_i \to E \mid i \in I\}$ in $\mathcal{E}$ and for each $i \in I$ a natural number n_i and an n_i-tuple $(\xi_1, \ldots, \xi_{n_i})$ of generalized elements $E_i \to MA$ such that $< \xi_1, \ldots, \xi_{n_i} >$ factors through the interpretation of R_{n_i} in M. By taking a_i to be the Σ-substructure of $\mathrm{Hom}_{\mathcal{E}}(E_i, M)$ on the finite subset consisting of the elements $\xi_1, \ldots, \xi_{n_i}$ and all the elements in the image of the homomorphism $\mathrm{Hom}_{\mathcal{E}}(e_i, M) \circ f$, we clearly obtain a structure satisfying the second axiom of $\mathbb{T}$ and also the last two groups of axioms of $\mathbb{T}$ (the validity of Horn sequents being inherited by substructures). Now, in order to obtain from this family of Σ-structure homomorphisms a 'localization' such that the domains of its homomorphisms satisfy all the axioms of $\mathbb{T}$, it suffices to 'localize' each f_i

a finite number of times (one for each (n_i+1)-tuple $\vec{u}$ of elements of a_i, where n_i is the cardinality of a_i), endowing the sets d_i arising in the surjection-inclusion factorizations of the homomorphisms with the quotient structure induced by the domains a_i via the relevant quotient map q_i (in the sense that the interpretation of each relation symbol R of Σ in such a set d_i is defined to be equal to the image of the interpretation of R in a_i under the quotient map q_i).

Since the Σ-structure homomorphisms $q_i : a_i \to d_i$ in such a 'localization' are quotient maps, the fact that the d_i satisfy the last two groups of axioms of $\mathbb{T}$ follows automatically from the fact that the a_i do. By the above considerations, it will thus be enough to show that each d_i satisfies the sequent σ_{n_i}, where n_i is the cardinality of d_iA, and the second axiom of $\mathbb{T}$. But the fact that each d_i satisfies the second axiom of $\mathbb{T}$ follows from the fact that a_i does. So the structures d_i satisfy all the axioms of $\mathbb{T}$.

This completes the proof that the hypotheses of Corollary 8.2.1 are satisfied by the theory $\mathbb{T}$ with respect to its Horn part and the category of (finite) $\mathbb{T}$-models; therefore $\mathbb{T}$ is of presheaf type classified by the topos $[\mathbf{S}, \mathbf{Set}]$ where $\mathbf{S}$ is the category of finite sets and surjective functions between them.

9.4 The theory of Diers fields

As observed in [51] (cf. section 5 therein), the coherent theory $\mathbb{T}$ of fields is not of presheaf type. In fact, one can easily identify two properties which are preserved by homomorphisms of fields but which are not definable in finitely generated fields by geometric formulae over the signature of $\mathbb{T}$ (cf. Theorem 6.1.4): the property of a field to have characteristic 0, and the property of a tuple $(x_1, \ldots, x_n, x_{n+1})$ of elements of a field that the element x_{n+1} is transcendental over the subfield generated by the elements $x_1, \ldots, x_n$. This can be easily seen by arguing as follows. Assuming the axiom of choice, the theory $\mathbb{T}$ has enough set-based models (it being coherent). Hence if the property of having characteristic 0 were definable by a geometric sentence ϕ over the signature of fields then the sequent $(\top \vdash_{[]} \phi \vee \bigvee_{p \in \mathbb{P}} \phi_p)$, where $\mathbb{P}$ is the set of prime numbers and ϕ_p (for each $p \in \mathbb{P}$) is the sentence $p.1 = 0$ expressing the property of having characteristic p, would be provable in $\mathbb{T}$. But, $\mathbb{T}$ being coherent, the sequent $(\top \vdash_{[]} \phi \vee \bigvee_{p \in \mathbb{P}'} \phi_p)$ would be provable in $\mathbb{T}$ for some finite subset $\mathbb{P}' \subseteq \mathbb{P}$ (cf. Theorem 10.8.6(iii) below), which is absurd as it would imply that the set of all possible characteristics of a field is finite. A similar argument works for the 'transcendence' property which, like the former, is the complement of a property definable by a strictly infinitary geometric formula.

In order to make such properties definable and obtain a presheaf completion of the theory $\mathbb{T}$, it is thus necessary to enlarge the signature of $\mathbb{T}$ with new relation symbols. In fact, Johnstone introduces in [51] a 0-ary predicate R_0, expressing the property of a field to have characteristic 0, and for each natural number $n \geq 0$ an $(n+1)$-predicate $R_{n+1}(x_1, \ldots, x_n, x_{n+1})$ expressing the property of x_{n+1} to be transcendental over the subfield generated by the elements $x_1, \ldots, x_n$. Let Σ'

be the resulting signature. Formally, one has to impose the following axioms over Σ' to ensure that these predicates have indeed the required meaning (below we use the abbreviation $\mathrm{Inv}(z)$ for the formula $(\exists x)(x \cdot z = 1)$):

$$\left(\top \vdash R_0 \vee \bigvee_{p \in \mathbb{P}} p\,.\,1 = 0\right);$$

$$(R_0 \wedge p\,.\,1 = 0 \vdash \bot)$$

for each $p \in \mathbb{P}$;

$$\Bigg(\top \vdash_{x_1,\ldots,x_{n+1}} R_n(x_1, \ldots, x_n, x_{n+1}) \vee \left(\bigvee_{m \in \mathbb{N},\ c_0^{\vec{n}},\ldots,c_m^{\vec{n}}} \left(\sum_{j=0}^{m} P_{c_j^{\vec{n}}} x_{n+1}^j = 0\right) \wedge \bigvee_{i \in \{0,1,\ldots,m\}} \mathrm{Inv}(P_{c_i^{\vec{n}}})\right)\Bigg)$$

for each natural number $n \geq 0$, where the former disjunction is taken over all the natural numbers $m \geq 0$ and all the tuples $c_i^{\vec{n}}$ (for $i \in \{0, \ldots, m\}$) of integer coefficients (i.e. coefficients of the form $1 + \cdots + 1$ an integer number of times) of polynomials in n variables $x_1, \ldots, x_n$ of degree $\leq m$ and the expression $P_{c_i^{\vec{n}}}$ (for each $i \in \{0, \ldots, m\}$) denotes the polynomial term in the variables $x_1, \ldots, x_n$ corresponding to the tuple $c_i^{\vec{n}}$, and

$$\Bigg(R_n(x_1, \ldots, x_n, x_{n+1}) \wedge \left(\bigvee_{m \in \mathbb{N},\ c_0^{\vec{n}},\ldots,c_m^{\vec{n}}} \left(\left(\sum_{j=0}^{m} P_{c_j^{\vec{n}}} x_{n+1}^j = 0\right) \wedge \bigvee_{i \in \{0,1,\ldots,m\}} \mathrm{Inv}(P_{c_i^{\vec{n}}})\right)\right) \vdash_{x_1,\ldots,x_{n+1}} \bot\Bigg).$$

Let $\mathbb{D}$ be the theory, called in [51] the *theory of Diers fields*, obtained from $\mathbb{T}$ by adding these new predicates and the above-mentioned axioms.

Note that the geometric formula

$$\bigvee_{m \in \mathbb{N},\ c_0^{\vec{n}},\ldots,c_m^{\vec{n}}} \left(\sum_{j=0}^{m} P_{c_j^{\vec{n}}} x_{n+1}^j = 0\right) \wedge \bigvee_{i \in \{0,1,\ldots,m\}} \mathrm{Inv}(P_{c_i^{\vec{n}}})$$

is $\mathbb{D}$-provably equivalent to a disjunction of geometric formulae, namely, the formulae $\left(\sum_{j=0}^{m} P_{c_j^{\vec{n}}} x_{n+1}^j = 0\right) \wedge \mathrm{Inv}(P_{c_i^{\vec{n}}})$ (for each m, $c_0^{\vec{n}}, \ldots, c_m^{\vec{n}}$ and $i \in \{0, \ldots, m\}$), each of which has a $\mathbb{D}$-provable complement, namely the formula

$$\mathrm{Inv}(\sum_{j=0}^{m} P_{c_j^{\vec{n}}} x_{n+1}^j) \vee P_{c_i^{\vec{n}}} = 0.$$

Notice also that, for any tuples $\vec{c_i^n}$ (for $i \in \{0, \ldots, m\}$) of integer coefficients of a polynomial expression $P_{\vec{c_i^n}}$ in the variables $x_1, \ldots, x_n$, the cartesian sequent

$$(*) \qquad \left(R_n(x_1, \ldots, x_n, x_{n+1}) \wedge \mathrm{Inv}(P_{\vec{c_i^n}}) \vdash_{x_1,\ldots,x_{n+1}} \mathrm{Inv}\left(\sum_{j=0}^{m} P_{\vec{c_j^n}} x_{n+1}^j \right) \right)$$

is provable in $\mathbb{D}$.

Following [51], we observe that the theory $\mathbb{D}$ satisfies the property that every finitely presentable $\mathbb{D}$-model, i.e. every finitely generated field, is finitely presented as a model of its cartesianization. This will follow from Lemma 7.1.7 once we have proved that every finitely generated field F is presented by a finite set of generators, in the weak sense of the lemma, by a formula over the signature of $\mathbb{D}$. To this end, we regard $\mathbb{T}$ as axiomatized over the signature of von Neumann regular rings, which contains a unary function for the operation of pseudoinverse; over this signature, every finitely generated field (in the sense of field theory) becomes finitely generated (in the sense of model theory, cf. section 7.2.3). Notice that $\mathbb{D}$ satisfies the first set of hypotheses of Lemma 7.1.7 with respect to the theory $\mathbb{T}$.

We shall prove that every field F with a finite set of generators is presented as a model of the cartesianization of $\mathbb{D}$, in the weak sense of Lemma 7.1.7, by a geometric formula over the signature of $\mathbb{D}$ by induction on the number n of generators of F. If $n = 0$ then F is equal to its prime field, so either F has characteristic p, in which case it is equal to $\mathbb{F}_p$, or F has characteristic 0, in which case it is equal to $\mathbb{Q}$. Now, the field $\mathbb{F}_p$ is clearly (weakly) presented by the formula $p \,.\, 1 = 0$, while the field $\mathbb{Q}$ is (weakly) presented by the formula R_0 since the sequent $(R_0 \vdash \mathrm{Inv}(n \,.\, 1))$ is provable in the cartesianization of $\mathbb{D}$ for each non-zero natural number n. Now, consider a field F generated by $n+1$ elements $x_1, \ldots, x_n, x_{n+1}$. If we denote by F_0 its prime field, we have that $F = F_0(x_1, \ldots, x_n)(x_{n+1})$. Suppose that $F_0(x_1, \ldots, x_n)$ is (weakly) presented by a formula in n variables $\phi(x_1, \ldots, x_n)$ with the elements $x_1, \ldots, x_n$ as generators. There are two cases: either the element x_{n+1} is transcendental over the field $F_0(x_1, \ldots, x_n)$ or not.

In the first case, F is isomorphic to the field of rational functions in one variable with coefficients in $F_0(x_1, \ldots, x_n)$, and it is (weakly) presented by the formula $\{x_1, \ldots, x_n, x_{n+1} \,.\, \phi \wedge R_{n+1}\}$; in other words, for any model $(A, \{AR_n \mid n \in \mathbb{N}\})$ of the cartesianization of $\mathbb{D}$, the function which assigns a ring homomorphism $f : F \to A$ with the property that $f(x_1, \ldots, x_n, x_{n+1}) \in AR_{n+1} \cap [[\vec{x} \,.\, \phi]]_A$ to the element $f(x_1, \ldots, x_n, x_{n+1})$ is injective and surjective on $AR_{n+1} \cap [[\vec{x} \,.\, \phi]]_A$. This can be proved as follows. Since F is generated by the elements $x_1, \ldots, x_{n+1}$, the injectivity is clear, so it remains to prove the surjectivity, i.e. that for any $(n+1)$-tuple $(a_1, \ldots, a_n, a_{n+1}) \in AR_{n+1} \cap [[\vec{x} \,.\, \phi]]_A$ there exists a ring homomorphism $F \to A$ which sends x_i to a_i for each $i \in \{1, \ldots, n+1\}$. By the induction hypothesis, since $(a_1, \ldots, a_n) \in [[\vec{x} \,.\, \phi]]_A$, there exists a unique ring homomorphism

$g : F_0(x_1, \ldots, x_n) \to A$ such that $g(x_i) = a_i$ for each $i \in \{1, \ldots, n\}$. By definition of F, there exists a ring homomorphism $f : F \to A$ which extends g and sends x_{n+1} to a_{n+1} if and only if for every polynomial P with a non-zero coefficient $P_{c_i^{\vec{n}}}(x_1, \ldots, x_n)$ in $F_0(x_1, \ldots, x_n)$, $\sum\limits_{j=0}^{m} g(P_{c_j^{\vec{n}}})a^j$ is invertible in A. But since $P_{c_i^{\vec{n}}}(x_1, \ldots, x_n)$ is non-zero (equivalently, invertible) in the field $F_0(x_1, \ldots, x_n)$, its image $g(P_{c_i^{\vec{n}}}(x_1, \ldots, x_n)) = P_{c_i^{\vec{n}}}(g(x_1), \ldots, g(x_n))$ under the homomorphism g is invertible in A; therefore, since the sequent $(*)$ holds in A, the condition $(f(x_1), \ldots, f(x_n), a) \in AR_{n+1}$ entails the fact that $\sum\limits_{j=0}^{m} g(P_{c_j^{\vec{n}}})a^j$ is invertible in A, as required.

In the second case, consider the minimal polynomial P for x_{n+1} over the field $F_0(x_1, \ldots, x_n)$; then F is isomorphic to the quotient of $F_0(x_1, \ldots, x_n)[X]$ by the ideal generated by the polynomial P. It is immediate to see that F is (weakly) presented by the formula in $n + 1$ variables $\phi \wedge P(x_{n+1}) = 0$ with generators $x_1, \ldots, x_n, x_{n+1}$.

These arguments show that the theories $\mathbb{T}$ and $\mathbb{D}$ satisfy the hypotheses of Lemma 7.1.7. It follows that all the finitely generated fields are finitely presented models of the cartesianization of $\mathbb{D}$, as required. Condition (iii) of Theorem 6.3.1 is therefore satisfied by $\mathbb{D}$ with respect to the category of finitely generated fields (cf. Proposition 6.3.12(i)). We could have alternatively deduced this from Corollary 6.3.17 and Theorem 6.3.22.

In [51], it is shown that $\mathbb{D}$ is a theory of presheaf type by assuming a form of the axiom of choice in order to ensure that $\mathbb{D}$ has enough set-based models. Theorem 6.3.1 allows us to prove that $\mathbb{D}$ is of presheaf type directly, without assuming any non-constructive principles. Having already proved that condition (iii) of Theorem 6.3.1 is satisfied, it remains to show that conditions (i) and (ii) hold as well. To prove this, we shall verify that conditions (ii)(a), (ii)(b) and (ii)(c) of Theorem 6.3.4 and conditions (ii)(a) and (ii)(b) of Theorem 6.3.8 are satisfied.

The fact that condition (ii)(a) of Theorem 6.3.8 holds follows immediately from the above discussion in view of Remarks 6.3.9(b) and 6.2.6(c). Condition (ii)(c) of Theorem 6.3.4 is trivially satisfied while condition (ii)(a) of Theorem 6.3.4 follows from condition (ii)(a) of Theorem 6.3.8 (cf. Remarks 6.3.5(b) and 6.3.5(c)). By Remark 6.3.9(a), condition (ii)(b) of Theorem 6.3.4 implies condition (ii)(b) of Theorem 6.3.8. So, to complete the proof that $\mathbb{D}$ is of presheaf type, it remains to show that condition (ii)(b) of Theorem 6.3.4 holds, i.e. that for any finitely generated fields c and d, any $\mathbb{D}$-model M in a Grothendieck topos $\mathcal{E}$ and any Σ'-structure homomorphisms $f : c \to \mathrm{Hom}_{\mathcal{E}}(E, M)$ and $g : d \to \mathrm{Hom}_{\mathcal{E}}(E, M)$, there exist an epimorphic family $\{e_i : E_i \to E \mid i \in I\}$ in $\mathcal{E}$ and for each $i \in I$ a finitely generated field b_i, field homomorphisms $u_i : c \to b_i$, $v_i : d \to b_i$ and a Σ'-structure homomorphism $z_i : b_i \to \mathrm{Hom}_{\mathcal{E}}(E_i, M)$ such that $\mathrm{Hom}_{\mathcal{E}}(e_i, M) \circ f = z_i \circ u_i$ and $\mathrm{Hom}_{\mathcal{E}}(e_i, M) \circ g = z_i \circ v_i$.

We can prove this by induction on the sum n of the minimal numbers of generators of c and d.

Before proceeding further, it is convenient to remark the following fact: for any model $(A, \{AR_n \mid n \in \mathbb{N}\})$ of the cartesianization of $\mathbb{D}$ where A is not the trivial ring (for instance, a Σ'-structure of the form $\mathrm{Hom}_{\mathcal{E}}(E, M)$, where M is a model of $\mathbb{D}$ in a topos $\mathcal{E}$ and E is a non-zero object of $\mathcal{E}$) and any field e considered as a model of $\mathbb{D}$, all the Σ'-structure homomorphisms $e \to A$ reflect the satisfaction of the relations R_n, i.e. $f(x_1, \ldots, x_n) \in AR_n$ implies $(x_1, \ldots, x_n) \in eR_n$. This easily follows from the disjunctive axioms of $\mathbb{D}$ defining R_n and the cartesian axiom $(*)$. Note also that such homomorphisms are always injective (since their domain is a field and their codomain is a non-zero ring).

For proving that our condition is satisfied, we can suppose without loss of generality all the objects E arising in Σ'-structure homomorphisms to structures of the form $\mathrm{Hom}_{\mathcal{E}}(E, M)$ to be non-zero (since removing zero arrows from an epimorphic family leaves the family epimorphic).

If $n = 0$ (that is, if both c and d are equal to their prime fields) then c and d have the same characteristic; indeed, equalities of the form $p.\,1 = 0$ are preserved and reflected by the homomorphisms f and g. So c and d are isomorphic, whence the condition is trivially satisfied. Let us now assume that the condition is true for all $k \leq n$ and prove it for $n + 1$. We can represent c as $c'(x)$ in such a way that a set of generators for c with minimal cardinality can be obtained by adding an element x to a set of generators for c'; then by the induction hypothesis there exist an epimorphic family $\{e_i : E_i \to E \mid i \in I\}$ in $\mathcal{E}$ and for each $i \in I$ a finitely generated field u_i, field homomorphisms $f_i : c' \to u_i$ and $g_i : d \to u_i$ and a Σ'-structure homomorphism $r_i : u_i \to \mathrm{Hom}_{\mathcal{E}}(E_i, M)$ such that $r_i \circ f_i = \mathrm{Hom}_{\mathcal{E}}(e_i, M) \circ f|_{c'}$ and $r_i \circ g_i = \mathrm{Hom}_{\mathcal{E}}(e_i, M) \circ g$. Now, consider for each $i \in I$ the element $f(x) \circ e_i \in \mathrm{Hom}_{\mathcal{E}}(E_i, M)$. Suppose that c' has m generators $\chi_1, \ldots, \chi_m$; then for each $i \in I$, by the disjunctive axiom of $\mathbb{D}$ involving R_m, there exists an epimorphic family $\{f_{i,j} : F_{i,j} \to E_i \mid j \in J_i\}$ in $\mathcal{E}$ such that for any $j \in J_i$, either $R_{m+1}((r_i \circ f_i)(\chi_1) \circ e_i \circ f_{i,j}, \ldots, (r_i \circ f_i)(\chi_m) \circ e_i \circ f_{i,j}, f(x) \circ e_i \circ f_{i,j})$ or $f(x) \circ e_i \circ f_{i,j}$ is the root of a non-zero polynomial P with coefficients belonging to the von Neumann regular subring of $\mathrm{Hom}_{\mathcal{E}}(F_{i,j}, M)$ generated by the elements $(r_i \circ f_i)(\chi_1) \circ e_i \circ f_{i,j}, \ldots, (r_i \circ f_i)(\chi_m) \circ e_i \circ f_{i,j}$. In the first case, $f(x) \circ e_i \circ f_{i,j}$ is transcendental over c' via the embedding $\mathrm{Hom}_{\mathcal{E}}(f_{i,j}, M) \circ r_i \circ f_i$ and over u_i via the embedding $\mathrm{Hom}_{\mathcal{E}}(f_{i,j}, M) \circ r_i$ (apply the above remarks to these two embeddings); it follows that the homomorphism f_i extends to a homomorphism from $c = c'(x)$ to u_i. In the second case, the homomorphism $\mathrm{Hom}_{\mathcal{E}}(e_i \circ f_{i,j}, M) \circ f$ being injective, there exists a non-zero polynomial P with coefficients in c' such that $f(x) \circ e_i \circ f_{i,j}$ is a root of the image of P under $\mathrm{Hom}_{\mathcal{E}}(f_{i,j}, M) \circ r_i \circ f_i = \mathrm{Hom}_{\mathcal{E}}(e_i \circ f_{i,j}, M) \circ f|_{c'}$. It follows that x is a root of P in c. We can thus clearly suppose P to be irreducible without loss of generality and represent $c = c'(x)$ as $c = c'[z]/P(z)$, via an isomorphism sending z to x; denoting by P' the image of P under the homomorphism f_i, the arrow f_i thus yields an arrow $c = c'[z]/P(z) \to u_i[w]/P'(w)$ and the homomorphism $\mathrm{Hom}_{\mathcal{E}}(f_{i,j}, M) \circ r_i$ factors through the quotient map $u_i \to u_i[w]/P'(w)$ yielding a ring homomorphism which is in fact a Σ'-structure homomorphism (by Lemma 7.1.7, cf. the above argument for proving that every finitely generated field is finitely presented as a model of the cartesianization of $\mathbb{D}$).

This completes our proof that $\mathbb{D}$ is of presheaf type.

Notice that we could have alternatively shown that $\mathbb{D}$ satisfies condition (ii)(b) of Theorem 6.3.4 either by using Remark 6.3.5(d) or by applying Theorem 6.3.6.

9.5 The theory of algebraic extensions of a given field

Let F be a field. We define the theory $\mathbb{T}_F$ of *algebraic extensions* of F as the expansion of the coherent theory of fields obtained by adding one constant symbol $\overline{a}$ for each element $a \in F$ and the following axioms:

$$(\top \vdash \overline{1_F} = 1);$$

$$(\top \vdash \overline{0_F} = 0);$$

$$(\top \vdash \overline{a} + \overline{b} = \overline{a +_F b})$$

for any elements $a, b \in F$ (where the symbol $+_F$ denotes the addition operation in F);

$$(\top \vdash \overline{a} \cdot \overline{b} = \overline{a \cdot_F b})$$

for any elements $a, b \in F$ (where the symbol $\cdot_F$ denotes the multiplication operation in F), plus the algebraicity axiom

$$\left(\top \vdash_x \bigvee_{n\in\mathbb{N},\ a_0,\dots,a_{n-1},a_n\in F} \overline{a_n}\cdot x^n + \overline{a_{n-1}}\cdot x^{n-1} + \cdots + \overline{a_0} = 0\right).$$

Let us prove that $\mathbb{T}_F$ is of presheaf type. Clearly, the finitely presentable models of $\mathbb{T}_F$ are exactly the finitely generated algebraic extensions of F, that is the finite extensions of F. One can prove, by adapting the argument used in the proof that every finitely generated field is finitely presented as a model of the cartesianization of the theory of Diers fields established in section 9.4, that every finite extension of F is finitely presented as a model of the cartesianization of $\mathbb{T}_F$. Specifically, every finite extension $F(x_1, \dots, x_n)$ of F is presented by the conjunction of the formulae of the form $P_i(x_1, \dots, x_i)(x_{i+1}) = 0$ (for $i = 0, \dots, n-1$), where P_i is the minimal polynomial of the element x_{i+1} over the field $F(x_1, \dots, x_i)$.

Condition (iii) of Theorem 6.3.1 is thus satisfied by the theory $\mathbb{T}_F$ with respect to the category of finite extensions of F (cf. Proposition 6.3.12(i)). In verifying that conditions (i) and (ii) of Theorem 6.3.1 are satisfied, one is reduced as in section 9.4 to check that condition (ii)(b) of Theorem 6.3.4 holds; again, this can be done by an argument similar to the one given in section 9.4. We can thus conclude that $\mathbb{T}_F$ is of presheaf type classified by the topos of covariant set-valued functors on the category of finite extensions of F.

Next, let us consider the theory $\mathbb{S}_F$ of $\mathbb{T}_F$ of *separable extensions* of F, that is the quotient of $\mathbb{T}_F$ obtained by adding the following sequent:

$$\left(\top \vdash_x \bigvee_{n\in\mathbb{N},\ (a_0,\ldots,a_{n-1},a_n)\in\mathcal{S}_F^n} \overline{a_n}\cdot x^n + \overline{a_{n-1}}\cdot x^{n-1} + \cdots + \overline{a_0} = 0\right),$$

where $\mathcal{S}_F^n$ is the set of $(n+1)$-tuples of elements $a_0, \ldots, a_n$ of F such that the polynomial $a_n Z^n + a_{n-1}Z^{n-1} + \cdots + a_0 \in F[Z]$ is irreducible and separable.

Clearly, the finitely presentable $\mathbb{S}_F$-models are precisely the finite separable extensions of F. In particular, every finitely presentable $\mathbb{S}_F$-model is finitely presentable also as a $\mathbb{T}_F$-model. In fact, by Artin's primitive element theorem, every finite separable extension of F is presented by a formula of the form $\{x\,.\,\overline{a_n}\cdot x^n + \overline{a_{n-1}}\cdot x^{n-1} + \cdots + \overline{a_0} = 0\}$.

The hypotheses of Corollary 8.2.1 are trivially satisfied by the quotient $\mathbb{S}_F$ of $\mathbb{T}_F$; indeed, for any $\mathbb{T}_F$-model homomorphism $M \to N$ (in an arbitrary Grothendieck topos), if N is separable (i.e. a model of $\mathbb{S}_F$) then M is *a fortiori* separable as well. We can thus conclude that the theory $\mathbb{S}_F$ is also of presheaf type, classified by the category of covariant set-valued functors on the category of finite separable extensions of F.

Remark 9.5.1 The theory of fields of fixed finite characteristic p which are algebraic over their prime field, which is shown in [29] to be of presheaf type classified by the topos of covariant set-valued functors on the category of finite fields of characteristic p, is (trivially) Morita-equivalent to the theory $\mathbb{T}_{\mathbb{F}_p}$ introduced above (where $\mathbb{F}_p$ is the field with p elements).

9.6 Groups with decidable equality

In this section we shall study the injectivization $\mathbb{G}_m$ of the (algebraic) theory $\mathbb{G}$ of groups and describe a presheaf completion for it (in the sense of section 7.1.3).

Clearly, the finitely presentable $\mathbb{G}_m$-models are precisely the finitely generated groups. Even if $\mathbb{G}_m$ satisfies condition (iii) of Theorem 6.3.1 with respect to the category of finitely generated groups and injective homomorphisms between them (by Theorem 6.3.22 and Corollary 6.3.17), $\mathbb{G}_m$ is not of presheaf type. To see this, consider the property of an element x of a group G to be non-nilpotent. This property is clearly preserved by injective homomorphisms of groups, so if $\mathbb{G}_m$ were of presheaf type it would be definable in finitely generated groups by a geometric formula $\phi(x)$ over the signature of $\mathbb{G}_m$ (by Theorem 6.1.4) and hence the sequent

$$\left(\top \vdash_x \phi(x) \vee \bigvee_{n\in\mathbb{N}} (x^n = 1)\right)$$

would be provable in $\mathbb{G}_m$ (since every theory of presheaf type has enough set-based models and this sequent is valid in every set-based $\mathbb{G}_m$-model by definition of ϕ). As $\mathbb{G}_m$ is coherent, the disjunction on the right-hand side could be replaced by a finite subdisjunction (cf. Theorem 10.8.6(iii)); but this is absurd since it would imply that there exists a natural number n such that every nilpotent element x of a group satisfies $x^n = 1$.

To obtain a presheaf completion of the theory $\mathbb{G}_m$, we add to the signature of $\mathbb{G}_m$ a relation symbol R^n_N for each natural number n and any normal subgroup N of the free group F_n on n generators, and the following axioms (where the symbol $\neq$ denotes the predicate of $\mathbb{G}_m$ which is $\mathbb{G}_m$-provably complemented to the equality relation):

$$\left(\top \vdash_{\vec{x}} R^n_N(\vec{x}) \vee \left(\bigvee_{w,w' \in F_n \mid ww'^{-1} \in N} w(\vec{x}) \neq w'(\vec{x}) \right) \vee \left(\bigvee_{w,w' \in F_n \mid ww'^{-1} \notin N} w(\vec{x}) = w'(\vec{x}) \right)\right)$$

and

$$\left(R^n_N(\vec{x}) \wedge \left(\bigvee_{w,w' \in F_n \mid ww'^{-1} \in N} w(\vec{x}) \neq w'(\vec{x}) \vee \bigvee_{w,w' \in F_n \mid ww'^{-1} \notin N} w(\vec{x}) = w'(\vec{x}) \right) \vdash_{\vec{x}} \bot \right)$$

for any $n \in \mathbb{N}$ and any normal subgroup N of F_n, and

$$\left(\top \vdash_{\vec{x}} \bigvee_{N \in \mathcal{N}_n} R^n_N(\vec{x})\right),$$

where $\mathcal{N}_n$ is the set of normal subgroups of F_n (for any $n \in \mathbb{N}$).

Let $\mathbb{G}_p$ be the resulting theory; we shall prove that it is of presheaf type.

The theories $\mathbb{G}_m$ and $\mathbb{G}_p$ clearly satisfy the first set of hypotheses of Lemma 7.1.7. Let us show that every finitely generated group is finitely presented as a model of the cartesianization of $\mathbb{G}_p$. By Lemma 7.1.7, to prove that F_n/N is presented by the formula $R^n_N(\vec{x})$ as a model of the cartesianization of $\mathbb{G}_p$, it suffices to verify that for any set-based model $(G', \{G'R^n_N \mid n \in \mathbb{N}, N \in \mathcal{N}_n\})$ of the cartesianization of $\mathbb{G}_p$, denoting by $\vec{\xi} = (\xi_1, \ldots, \xi_n)$ the generators of the group F_n, the group homomorphisms $f : F_n/N \to G'$ such that $f(\vec{\xi}) \in [[\vec{x} \,.\, R^n_N]]_{G'}$ correspond exactly, via the assignment $f \to f(\vec{\xi})$, to the n-tuples $\vec{y}$ of elements of G' which belong to the interpretation in G' of the formula R^n_N. Given an n-tuple $\vec{y}$ of elements of G' which belong to the interpretation of R^n_N in G', we can define a function $f : F_n/N \to G'$ by setting $f([w]) = w_{G'}(\vec{y})$ (where $w_{G'}$ is the interpretation in G' of the term $w \in F_n$). This is a well-defined injective group homomorphism since the following sequents are provable in the cartesianization of $\mathbb{G}_p$ and hence are valid in G':

$$(R^n_N \vdash_{\vec{x}} w = w')$$

for any $w, w' \in F_n$ such that $ww'^{-1} \in N$, and

$$(R^n_N \vdash_{\vec{x}} w \neq w')$$

for any $w, w' \in F_n$ such that $ww'^{-1} \notin N$. Since every finitely generated group is, up to isomorphism, of the form F_n/N for some natural number n and some normal subgroup N of F_n, we can conclude that the theory $\mathbb{G}_p$ satisfies condition (iii) of Theorem 6.3.1 with respect to the category $\mathbf{Grp}_{\text{f.g.}}$ of finitely generated groups and injective homomorphisms between them (cf. Proposition 6.3.12(i)). We could have alternatively proved this by invoking Corollary 6.3.17 and Theorem 6.3.22.

As in the case of the theory of Diers fields treated in section 9.4, in order to verify that the theory $\mathbb{G}_p$ satisfies conditions (i) and (ii) of Theorem 6.3.1 one is reduced to show that it satisfies condition (ii)(b) of Theorem 6.3.4; but this follows from Remark 6.3.5(d). The theory $\mathbb{G}_p$ is thus classified by the topos $[\mathbf{Grp}_{\text{f.g.}}, \mathbf{Set}]$.

Notice that the category $\mathbf{Grp}_{\text{f.g.}}$ is cocartesian; indeed, it has an initial object (namely, the trivial group) and pushouts (given by the free product with amalgamation construction, cf. [69]). It follows that the topos $[\mathbf{Grp}_{\text{f.g.}}, \mathbf{Set}]$ is coherent. The theory $\mathbb{G}_p$, in spite of being infinitary, is thus classified by a coherent topos. This fact has various implications for $\mathbb{G}_p$. For instance, the coherence of the classifying topos $\mathbf{Set}[\mathbb{G}_p]$ of $\mathbb{G}_p$ implies that every formula over the signature of $\mathbb{G}_p$ presenting a finitely generated group (regarded as a $\mathbb{G}_p$-model) is not only $\mathbb{G}_p$-irreducible, but also a coherent object of $\mathbf{Set}[\mathbb{G}_p]$; in particular, for any $\mathbb{G}_p$-compact formula $\{\vec{x}\,.\,\phi\}$ over the signature of $\mathbb{G}_p$, the formula $\{\vec{x}, \vec{y}\,.\,\phi(\vec{x}) \wedge \phi(\vec{y})\}$, where $\vec{x}$ and $\vec{y}$ are two disjoint contexts of the same type, is also $\mathbb{G}_p$-compact (cf. Definition 10.8.4(b) below for the notion of $\mathbb{T}$-compact formula for a geometric theory $\mathbb{T}$). Semantically speaking, a formula $\{\vec{x}\,.\,\phi\}$ is $\mathbb{G}_p$-compact if whenever $\{S_i \mid i \in I\}$ is a family of assignments $G \to S_i^G \subseteq G^n$ sending each finitely generated group G to a subset $S_i^G \subseteq [[\vec{x}\,.\,\phi]]_G$ in such a way that every injective homomorphism $f : G \to G'$ of finitely generated groups sends tuples in S_i^G to tuples in $S_i^{G'}$, if $[[\vec{x}\,.\,\phi]]_G = \bigcup_{i \in I} S_i^G$ then there exists a finite subset $J \subseteq I$ such that $[[\vec{x}\,.\,\phi]]_G = \bigcup_{i \in J} S_i^G$ for all G. Assuming the axiom of choice, by Deligne's theorem (Theorem 2.1.12) the fact that the category $\mathbf{Grp}_{\text{f.g.}}$ is cocartesian also implies the existence of enough set-based models for any quotient of $\mathbb{G}_p$ whose associated Grothendieck topology on $(\mathbf{Grp}_{\text{f.g.}})^{\text{op}}$ is of finite type.

9.7 Locally finite groups

Let $\mathbb{G}$ be the algebraic theory of groups. Since every finite group is finitely presented as a $\mathbb{G}$-model (cf. Proposition 9.1.1) and $\mathbb{G}$ is of presheaf type, Theorem

8.2.10 ensures that the quotient $\mathbb{U}$ of $\mathbb{G}$ consisting of all the geometric sequents over the signature of $\mathbb{G}$ that are valid in every finite group is classified by the topos $[\mathbf{Grp}_{\text{fin}}, \mathbf{Set}]$, where $\mathbf{Grp}_{\text{fin}}$ is the category of finite groups and homomorphisms between them. Since $\mathbb{U}$ is classified by the topos $[\mathbf{Grp}_{\text{fin}}, \mathbf{Set}]$, the set-based models of $\mathbb{U}$ are exactly the groups which can be expressed as filtered colimits of finite groups. This yields the following characterization of locally finite groups (recall that a group is said to be *locally finite* if all its finitely generated subgroups are finite).

Proposition 9.7.1 *The locally finite groups are exactly the groups which satisfy all the geometric sequents over the signature of the (algebraic) theory of groups which hold in all finite groups.*

Proof In view of the above discussion, it remains to verify that a group is locally finite if and only if it is a filtered colimit of finite groups. This can be proved as follows. If a group is locally finite then it is the filtered union of all its finitely generated (whence finite) subgroups. Conversely, suppose that G is a filtered colimit of finite groups. Then G is the directed union of the images in G of these finite groups, which are again finite. It follows that every finitely generated subgroup H of G is contained in one of the groups in this directed union (since the colimit is filtered, there exists one of them which contains all the generators of H) and hence it is *a fortiori* finite, as required. □

The injectivization of $\mathbb{U}$ is also of presheaf type (by Corollary 7.2.43) and can be characterized as the quotient of the injectivization $\mathbb{G}_m$ of $\mathbb{G}$ consisting of all the geometric sequents over its signature which hold in all finite groups.

9.8 Vector spaces

Let $\mathbb{V}_K$ be the expansion of the algebraic theory of vector spaces over a field K obtained by adding, for each natural number n, an n-ary predicate R_n expressing the property of an n-tuple of elements to be linearly independent, that is the following sequents:

$$\left(\top \vdash_{\vec{x}} R_n(\vec{x}) \vee \bigvee_{\substack{(k_1,\ldots,k_n)\in K^n \\ k_i\neq 0 \text{ for some } i}} k_1x_1 + \cdots + k_nx_n = 0 \right)$$

and

$$\left(R_n(\vec{x}) \wedge \left(\bigvee_{\substack{(k_1,\ldots,k_n)\in K^n \\ k_i\neq 0 \text{ for some } i}} k_1x_1 + \cdots + k_nx_n = 0 \right) \vdash_{\vec{x}} \bot \right).$$

The category of set-based models of the theory $\mathbb{V}_K$ has as objects the vector spaces over K and as arrows the injective homomorphisms between them. The finitely presentable $\mathbb{V}_K$-models are precisely the finite-dimensional vector spaces over K.

By using techniques analogous to those employed in section 9.4, one can prove that $\mathbb{V}_K$ is of presheaf type. Also, by using Corollary 8.2.1, one can easily show that, for any fixed natural number n, the expansion of the theory $\mathbb{V}_K$ obtained by adding the sequent

$$\left(\top \vdash_{x_1,\ldots,x_{n+1}} \bigvee_{\substack{(k_1,\ldots,k_{n+1})\in K^{n+1} \\ k_i\neq 0 \text{ for some } i}} k_1x_1 + \cdots + k_{n+1}x_{n+1} = 0 \right)$$

is of presheaf type; note that the set-based models of this theory are precisely the vector spaces over K of dimension $\leq n$.

9.9 The theory of abelian ℓ-groups with strong unit

Recall that an abelian ℓ-group with strong unit is a lattice-ordered group $(G, 0, \leq)$ with a distinguished element u, called the unit of the group, such that for any element $x \in G$ such that $x \geq 0$ there exists a natural number n such that $x \leq nu$, where $nu = u + \cdots + u$ n times. The reader may consult Chapter 2 of [70] for an introduction to the theory of lattice-ordered groups (also called ℓ-groups).

We can axiomatize the theory $\mathbb{L}_u$ of abelian ℓ-groups with strong unit over a signature Σ consisting of four binary function symbols $+$, $-$, inf, sup, two constants 0 and u and a binary relation symbol $\leq$, by using Horn sequents to formalize the notion of abelian ℓ-group and the following geometric sequent to express the property of being a strong unit:

$$\left(x \geq 0 \vdash_x \bigvee_{n\in\mathbb{N}} x \leq nu \right).$$

The following lemma will be useful for proving that the theory $\mathbb{L}_u$ is of presheaf type.

Lemma 9.9.1 *Let G be an abelian group with a distinguished element u and generators $x_1, \ldots, x_n$. If for every $i \in \{1, \ldots, n\}$ there exists a natural number k_i such that $|x_i| \leq k_i u$ then u is a strong unit for G.*

Proof Recall that the absolute value $|x|$ of an element x of an abelian ℓ-group $(G, 0, +, -, \leq, \text{inf}, \text{sup})$ is the element $\text{sup}(x, -x)$. For any $x \in G$, $|x| \geq 0$, $|x| = |-x|$, and for any $x, y \in G$ the triangular inequality $|x + y| \leq |x| + |y|$ holds.

Since G is generated by elements $x_1, \ldots, x_n$, every element x of G can be expressed as the interpretation $t(x_1, \ldots, x_n)$ of a term t over the signature Σ. Let us prove that there exists a natural number n such that $|x| \leq nu$ by induction on the structure of t; this will clearly imply our thesis, since if $x \geq 0$ then $|x| = x$. If t is a variable then the claim is clearly true by our hypotheses. If $x = x' + x''$ with $|x'| \leq n'u$ and $|x''| \leq n''u$ then by the triangular inequality we have $|x| \leq n' + n''$, and similarly for the subtraction, inf and sup cases. □

Let us verify that the theory $\mathbb{L}_u$ satisfies the hypotheses of Corollary 8.2.1 with respect to its Horn part (i.e. the theory $\mathbb{H}$ consisting of all the Horn sequents which are provable in $\mathbb{L}_u$).

We have to prove that for any finitely presentable $\mathbb{H}$-model c, any model G of $\mathbb{L}_u$ in a Grothendieck topos $\mathcal{E}$, any object E of $\mathcal{E}$ and any Σ-structure homomorphism $f : c \to \mathrm{Hom}_{\mathcal{E}}(E, G)$, there exist an epimorphic family $\{e_i : E_i \to E \mid i \in I\}$ in $\mathcal{E}$ and for each $i \in I$ a finitely presentable model c_i of $\mathbb{L}_u$ and Σ-structure homomorphisms $f_i : c \to c_i$ and $u_i : c_i \to \mathrm{Hom}_{\mathcal{E}}(E_i, G)$ such that $\mathrm{Hom}_{\mathcal{E}}(e_i, G) \circ f = u_i \circ f_i$. Let us suppose that c is presented as a $\mathbb{H}$-model by a cartesian formula $\phi(x_1, \ldots, x_n)$ with generators $\xi_1, \ldots, \xi_n$. Since G is an ℓ-group with strong unit, there exist an epimorphic family $\{e_i : E_i \to E \mid i \in I\}$ in $\mathcal{E}$ and for each $k \in \{1, \ldots, n\}$ and $i \in I$ a natural number $m_{k,i}$ such that $f(|\xi_k|) \circ e_i \leq m_{k,i} u_{\mathrm{Hom}_{\mathcal{E}}(E_i, G)}$ (where $u_{\mathrm{Hom}_{\mathcal{E}}(E_i,G)}$ denotes the unit of the ℓ-group $\mathrm{Hom}_{\mathcal{E}}(E_i, G)$). For each $i \in I$, let c_i be the $\mathbb{H}$-model presented by the cartesian formula $\phi(x_1, \ldots, x_n) \wedge |x_1| \leq m_{1,i} \wedge \cdots \wedge |x_n| \leq m_{n,i}$. For each $i \in I$, we have a natural quotient homomorphism $f_i : c \to c_i$ through which $\mathrm{Hom}_{\mathcal{E}}(e_i, G) \circ f$ factors, and the resulting factorization u_i satisfies the required property that $\mathrm{Hom}_{\mathcal{E}}(e_i, G) \circ f = u_i \circ f_i$. Since $\mathbb{H}$ is a Horn theory, each c_i is generated by the n-tuple $(f_i(\xi_1), \ldots, f_i(\xi_n))$ which presents it as a $\mathbb{H}$-model (cf. Remark 6.1.18(a)). Therefore the c_i are ℓ-groups with strong unit by Lemma 9.9.1. This argument also shows that the finitely presentable $\mathbb{L}_u$-models are exactly the finitely presented $\mathbb{H}$-models whose unit is strong (cf. Theorem 8.2.6).

Therefore all the hypotheses of Corollary 8.2.1 are satisfied and we can conclude that the theory $\mathbb{L}_u$ is of presheaf type. In fact, $\mathbb{L}_u$ is shown in [30] to be Morita-equivalent to the (algebraic) theory of MV-algebras.